

## WEEK 5 Class Exercise for Logistic Regression

**Data:** banknote\_authentication.csv

**Target:** forged – a binary target

**ML Objective:** Identify forged checks based upon an analysis of their signatures.

Data were extracted from 1,372 images of genuine and forged banknote-like specimens. For digitization, an industrial camera designed for print inspection was used.

The final images are 400x 400 pixels. Due to the object lens and distance to the investigated object, gray-scale pictures with a resolution of about 660 dpi were gained. Wavelet Transform tool were used to extract features from images.

| Digital Label | Role   | Level    | Description                  |
|---------------|--------|----------|------------------------------|
| forged        | Target | Binary   | 0=not forged, 1=forged check |
| variance      | Input  | Interval | -8 to 8                      |
| skewness      | Input  | Interval | -14 to 14                    |
| kurtosis      | Input  | Interval | -6 to 18                     |
| entropy       | Input  | Interval | -9 to 3                      |

**Note:** source *Ucl Machine Learning Repository*.

This exercise is to build a ML model (logistic regression) to automatically identify forged checks based upon the statistics calculated by comparing their bank signature with the signature on the checks. There are four measurements of this difference, all Interval measurements.

Refer to the Cursor Project Checklist distributed this week.